

Reproducing an ICA–FNC Framework for Schizophrenia Classification Using the COBRE Resting-State fMRI Dataset

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1. Introduction

Schizophrenia is a severe psychiatric illness, with abnormalities in perception, cognition, emotion, and behavior (DSM-5). More and more studies have shown that schizophrenia is not only related to the dysfunction of isolated brain regions, but reflects widespread disruptions in communication among distributed neural systems. This view is often called the dysconnection hypothesis (Friston & Frith, 1995; Stephan et al., 2006; Friston et al., 2016). The core idea of this view is that impaired integration among large-scale brain networks plays a central role in the pathophysiology of the disorder.

Resting-state functional magnetic resonance imaging (rs-fMRI) has become a key research tool for research on large-scale brain organization. Unlike task-based paradigms, rs-fMRI doesn't require specific behavioral tasks and can evaluate the activity of many different functional systems at once (Raichle et al., 2001; Fox & Raichle, 2007). Therefore, resting-state methods are increasingly becoming more important in psychiatric research areas including schizophrenia.

One of the most frequently used analysis methods for rs-fMRI data is Independent Component Analysis (ICA). Spatial ICA can separate fMRI signals into spatially independent functional networks and associated time courses. It doesn't require the definition of seed regions to achieve data-driven identification of brain networks (Beckmann & Smith, 2004; Calhoun et al., 2001). Because ICA can identify networks such as Default Mode Network (DMN), Sensorimotor Network, Visual Network, and Frontoparietal Network, it has become a standard approach for studying functional organization in schizophrenia (Damoiseaux et al., 2006).

On the basis of ICA, Functional Network Connectivity (FNC) further quantifies temporal correlations among ICA-derived networks. Distinct from methods which analyze the connectivity between single voxels or regions of interest (ROI), FNC focuses on the interactions among the entire functional systems, and can therefore describe brain organization from a higher level (Jafri et al., 2008). Past FNC research have noted that schizophrenia patients have wide-spread large-scale brain network interaction abnormalities, including the default mode, frontal, temporal, visual, and sensorimotor networks (Jafri et al., 2008; Damaraju et al., 2014).

A seminal study published by Arbabshirani et al. in 2013 further shows that FNC features extracted from resting-state ICA networks can distinguish schizophrenia patients from healthy controls using machine-learning classifiers. This study achieves relatively high classification performance and notes that abnormalities in large-scale functional integration contain diagnostically relevant information.

This project aims to use the publicly available COBRE dataset to reproduce and extend the classic (Arbabshirani et al., 2013) ICA-FNC classification framework. This study uses group ICA to identify resting state networks, uses dual regression to reconstruct subject specific network time-courses, and computes pairwise functional network connectivity measures and evaluates their ability to distinguish schizophrenia patients from healthy controls. In addition to classification performance, the project examines group differences in network connectivity patterns to identify candidate network interactions associated with schizophrenia. We hypothesized that schizophrenia patients would exhibit altered functional network connectivity, particularly involving default mode and sensorimotor systems, and that these abnormalities would contribute to successful classification of diagnostic status.

2. Methods

2.1 Dataset

We obtained resting-state functional magnetic resonance imaging (rs-fMRI) data from the Center for Biomedical Research Excellence (COBRE) (Aine et al., 2017) dataset. We used the NIAK lightweight preprocessed release (Bellec et al., 2012), which provides standardized preprocessing across participants.

A total of 145 participants were included in the current analysis, consisting of 71 schizophrenia patients and 74 healthy controls.

We acquired functional images using a gradient-echo echo-planar imaging (EPI) sequence with the following parameters (Aine et al., 2017):

- repetition time (TR) = 2 s,
- echo time (TE) = 29 ms,
- flip angle = 75°,
- voxel size = 3 × 3 × 4 mm³,
- matrix size = 64 × 64,
- number of slices = 32,
- number of volumes = 150.

Structural and demographic information associated with the COBRE dataset were obtained from the accompanying phenotypic files.

2.2 Preprocessing

In order to reduce the effect of non-neuronal noise on functional connectivity analysis, this study conducted additional pre-processing. The head-motion model is built by six rigid-body motion parameters corresponding to translations and rotations along the three spatial axes (Friston et al., 1996). To reduce physiological noise unrelated to neuronal activity, the average signal coming from white matter and the ventricular/CSF are regressed (Weissenbacher et al., 2009); also, when the data is viable, the study included the first five principal components of CompCor to account for additional physiological fluctuations (Behzadi et al., 2007). Finally, the study conducts linear detrending, applies temporal high-pass filtering (cut-off = 0.008 Hz) to the time series of every voxel in order to detach slow scanner drift while preserving resting-state fluctuations (Cordes et al., 2001). Finally, missing values are substituted by 0 to ensure the stability of later analysis.

2.3 Brain Mask Generation

This study then constructs a common group brain-mask based on the standardized MNI152 template. In order to ensure the quality of spatial resolution of the functional image while lowering computational cost, the study uses the 6 mm isotropic template. Using Nilearn's (Abraham et al., 2014) EPI mask generation procedure, the study estimates a binary brain mask. A common brain mask was used to ensure that voxel locations were spatially aligned across participants.

2.4 Time Series Extraction

Here, NiftiMasker in Nilearn (Abraham et al., 2014) is used to extract the voxel-wise functional time series. For every participant, 4D rs-fMRI data is transformed into a 2D matrix, where its dimension is 150 time points x 8580 brain voxels. Then, voxel time series were standardized to zero mean and unit variance. Each participant's data matrix is further concatenated along the temporal dimension, in order to build the group data matrix. Because every participant provides 150 time points, the final group (training set) matrix dimension becomes: (15150, 8580) (the training set takes 101 participants, with 52 controls and 49 patients). This temporal concatenation approach follows the classical Group ICA framework proposed by Calhoun et al. (2001) and has been widely adopted in group-level resting-state fMRI analyses.

2.5 Group Independent Component Analysis

2.5.1 Group Principal Component Analysis

Prior to ICA decomposition, dimensionality reduction was performed using Principal Component Analysis (PCA). Voxel activities are highly correlated and contain substantial redundancy. PCA identifies orthogonal directions that maximize variance in the data, providing a compact representation while preserving dominant large-scale patterns of brain activity (Jolliffe, 2002). In

group ICA frameworks, PCA is routinely applied before ICA to reduce computational complexity and suppress noise while retaining the most informative variance structure of the data (Calhoun et al., 2009).

A total of 120 principal components were retained. The cumulative explained variance spectrum was inspected to verify that the retained components captured a substantial proportion of the group-level variability.

2.5.2 Independent Component Analysis

Independent Component Analysis (ICA) was performed using the FastICA algorithm implemented in Scikit-learn. ICA assumes that the observed fMRI signals can be represented as linear mixtures of statistically independent latent sources (Hyvärinen & Oja, 2000):

$$X=AS$$

Where X denotes the observed data matrix, A is the mixing matrix, and S contains the latent independent components.

The FastICA algorithm estimates the unmixing matrix by maximizing non-Gaussianity, enabling efficient recovery of statistically independent spatial patterns from high-dimensional neuroimaging data (Hyvärinen, 1999). ICA has become one of the most widely used data-driven approaches for identifying resting-state functional networks from fMRI data (Calhoun et al., 2001).

The following parameters were used: n_components = 25, max_iter = 1000, tolerance = 10^{-4} , and random_state = 42. A total of 25 independent components were estimated.

2.5.3 Reconstruction of Group Spatial Maps

FastICA produces both a source matrix and a mixing matrix. To recover spatial brain maps, ICA components were projected back into voxel space by group spatial maps = $A^T P$ (Calhoun et al., 2001; Calhoun et al., 2009), where P denotes the PCA loading matrix.

Spatial maps were standardized using voxel-wise z-scores:

$$z=(x-\mu)/\sigma$$

The standardized component maps were transformed back into three-dimensional brain images using the inverse transformation of the NiftiMasker (Abraham et al., 2014).

2.6 Identification of Resting-State Networks

Group independent components were compared with canonical resting-state network templates derived from the Yeo seven-network atlas (Yeo et al., 2011).

Spatial correlations were computed between each independent component and the following canonical networks:

- Visual,
- Somatomotor,
- Dorsal Attention,
- Ventral Attention/Salience,
- Limbic,
- Frontoparietal,
- Default Mode.

This procedure follows common practices in resting-state ICA studies that identify biologically meaningful intrinsic connectivity networks through spatial correspondence with established network templates (Allen et al., 2011).

Components showing strong correspondence with canonical resting-state systems were retained for further analyses, whereas components dominated by artifacts were excluded.

The final analysis retained 10 components.

2.7 Dual Regression

Subject-specific representations of the group independent components were estimated using dual regression.

First, the group-level spatial maps obtained from Group ICA were regressed onto each participant's preprocessed fMRI data to estimate subject-specific component time courses.

Second, the estimated component time courses were regressed back onto the participant's fMRI data to obtain subject-specific spatial maps (Arbabshirani et al., 2013).

This procedure generated participant-specific estimates of both component time courses and spatial maps while preserving correspondence with the group-level ICA components.

2.8 Functional Network Connectivity

Functional Network Connectivity (FNC) was estimated using subject-specific component time courses (Jafri et al., 2008).

For each participant, Pearson correlation coefficients were calculated between all pairs of retained independent components.

Correlation coefficients were transformed using Fisher's z transformation:

$$z = \frac{1}{2} \ln\left[\frac{(1+r)}{(1-r)}\right].$$

The resulting FNC matrices quantified interactions among large-scale resting-state networks.

Upper triangular entries of the FNC matrices were extracted to generate feature vectors for subsequent analyses.

2.9 Statistical Analysis

Group differences between schizophrenia patients and healthy controls were assessed for each FNC feature. Statistical comparisons were performed using independent two-sample Welch's t-tests (Welch, 1947). Correction for multiple comparisons was performed using the False Discovery Rate (FDR) procedure with a significance threshold of $q < 0.05$ (Benjamini & Hochberg, 1995). Effect sizes were quantified using Cohen's d (Cohen, 1988).

2.10 Machine Learning Classification

Functional Network Connectivity features were used for diagnostic classification.

Participants were divided into independent training and testing datasets using a (101/44) split.

To avoid information leakage, all model fitting procedures, including dimensionality reduction and feature selection, were performed exclusively within the training data.

A Logistic Regression classifier was trained to distinguish schizophrenia patients from healthy controls.

Model performance was evaluated using:

- Accuracy,
- Sensitivity,
- Specificity,
- Area Under the ROC Curve (AUC).

Statistical significance was assessed using permutation testing (Ojala M & Garriga GC, 2010) with 1000 random label permutations and the ROC curve is as computed.

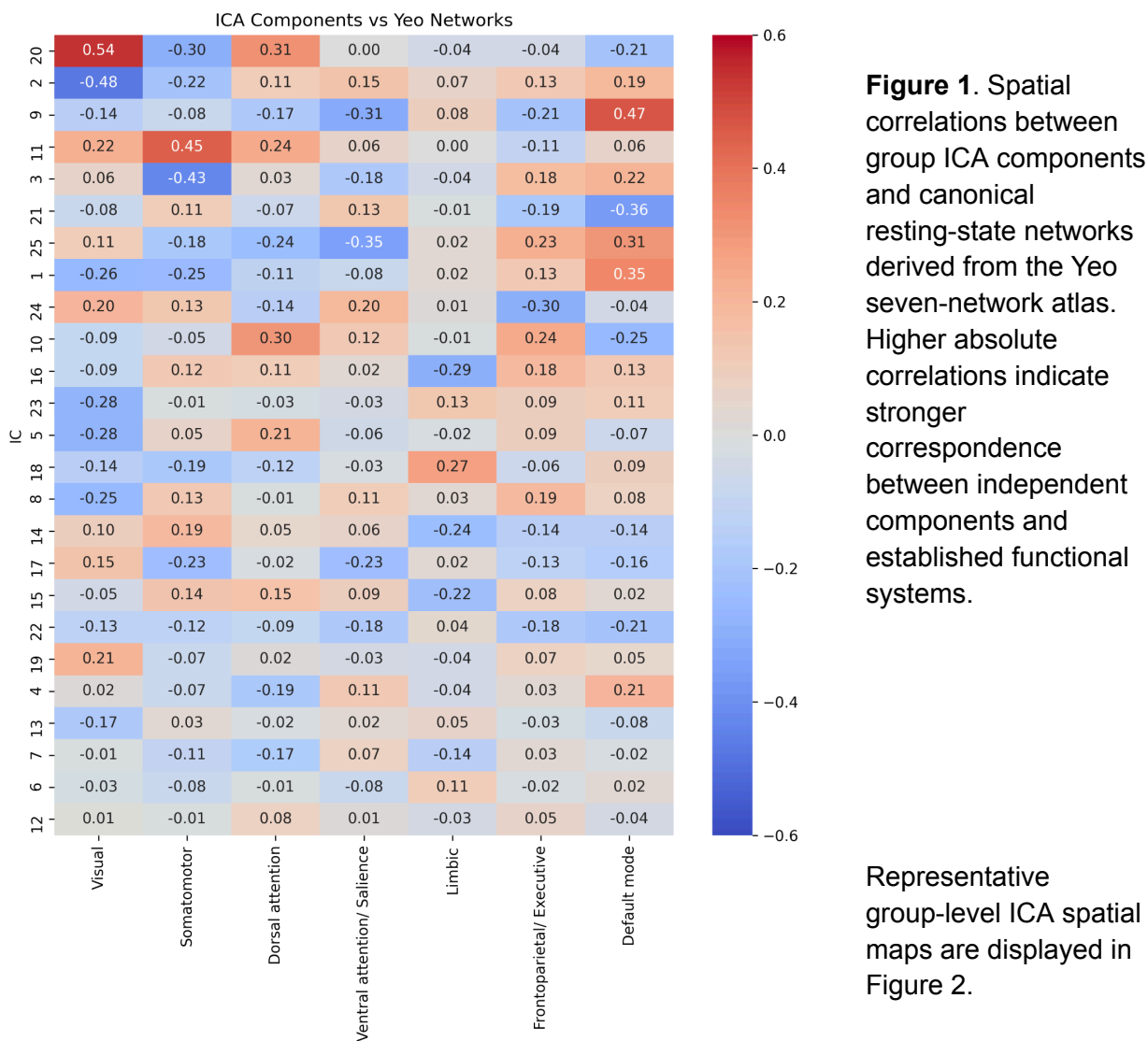
2.11 Software

All analyses were performed in Python 3.8 using Nilearn (Abraham et al., 2014), NiBabel (Brett et al., 2023), Scikit-learn (Pedregosa et al., 2011), SciPy (Virtanen et al., 2020), NumPy (Harris et al., 2020), and Matplotlib (Hunter, 2007).

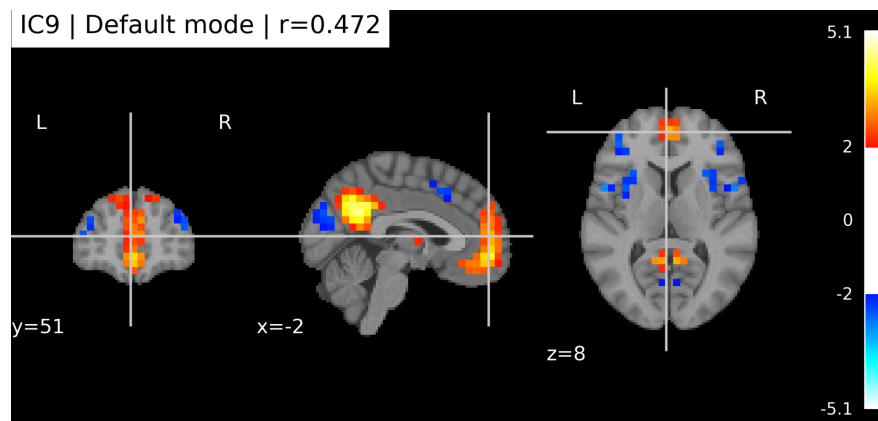
3. Results

3.1 Identification of Resting-State Networks

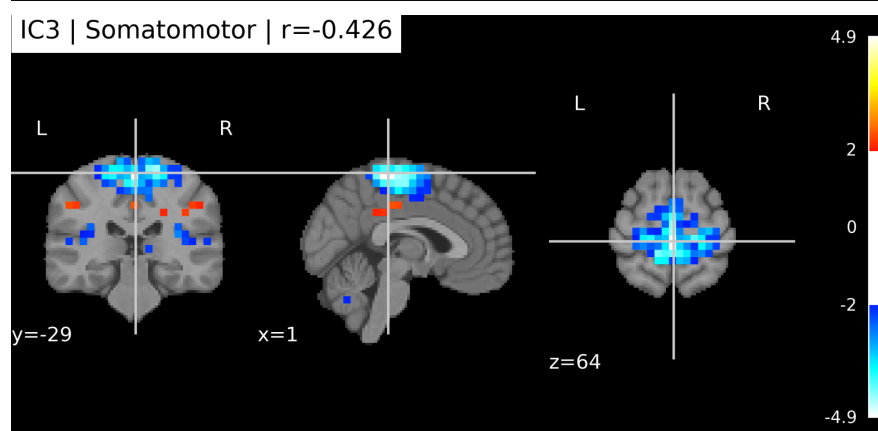
Group ICA identified 25 independent components. Spatial correlations with the Yeo seven-network atlas revealed components corresponding to canonical resting-state systems, including the Default Mode Network, Somatomotor Network, Visual Network, Frontoparietal/Executive Network, Dorsal Attention Network, Ventral Attention/Salience, and Limbic Network (Figure 1). Ten components showing clear correspondence with established resting-state networks were retained for subsequent analyses.



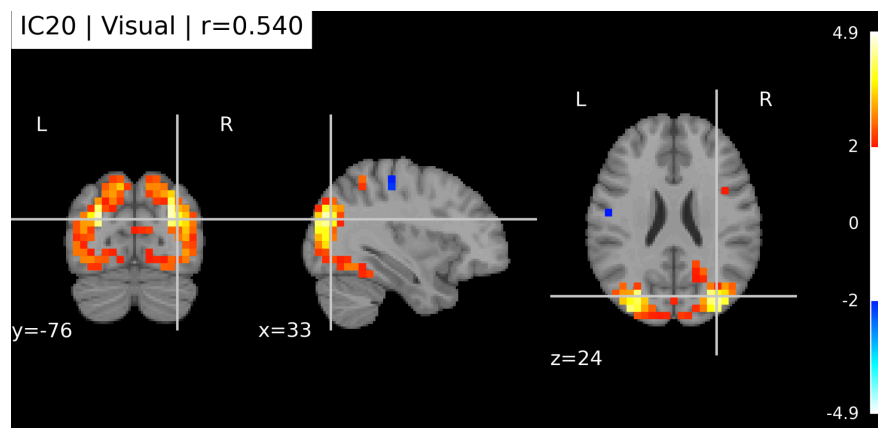
IC9 | Default mode | $r=0.472$



IC3 | Somatomotor | $r=-0.426$



IC20 | Visual | $r=0.540$



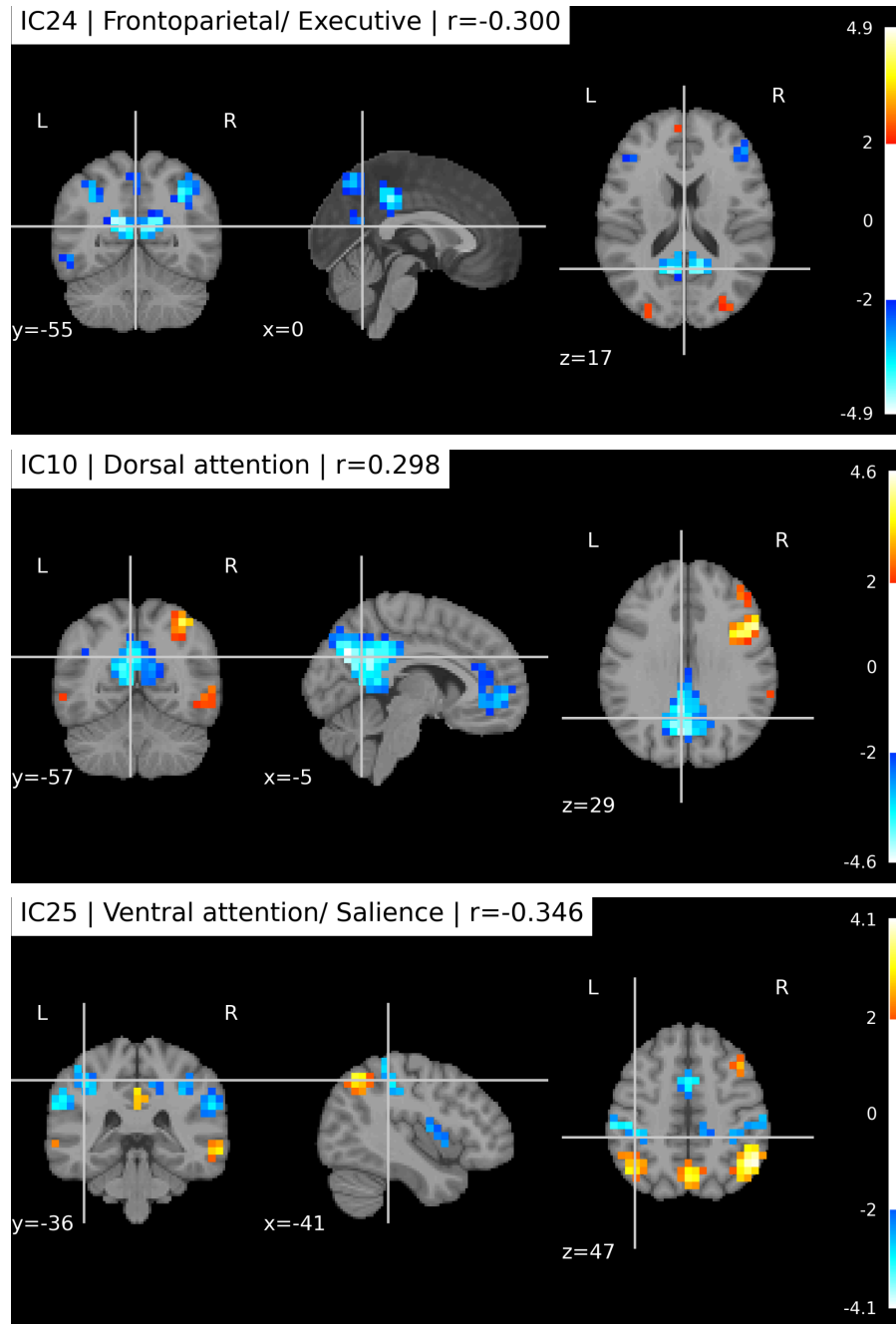


Figure 2. Representative group-level ICA spatial maps retained for subsequent analyses.

Displayed components show correspondence with the Default Mode Network (IC9), Somatomotor Network (IC3), Visual Network (IC20), Frontoparietal/Executive Network (IC24), Dorsal Attention Network (IC10), and Ventral Attention/Salience Network (IC25).

Positive and negative z-score values indicate regions contributing most strongly to each independent component.

	Feature	Network Edge	t	p	Cohen_d	HC mean	SZ mean	Difference	p_fdr	Significant
18	IC9-IC3	Default mode - Somatomotor	3.699665	0.000359	0.738925	0.088090	-0.072667	-0.160757	0.016165	True
17	IC9-IC11	Default mode - Somatomotor	2.262594	0.025887	0.451513	-0.098096	-0.227860	-0.129764	0.388304	False
9	IC2-IC9	Visual - Default mode	2.275068	0.025059	0.452426	0.000421	-0.099427	-0.099848	0.388304	False
7	IC20-IC24	Visual - Frontoparietal/Executive	2.016001	0.046543	0.401996	0.054974	-0.032560	-0.087533	0.523610	False
21	IC9-IC1	Default mode - Default mode	-1.781195	0.078065	-0.356078	-0.054793	0.019622	0.074415	0.585485	False
39	IC25-IC1	Ventral attention/Saliency - Default mode	1.801817	0.074751	0.360349	0.001018	-0.081153	-0.082171	0.585485	False
34	IC3-IC10	Somatomotor - Dorsal attention	1.626832	0.106999	0.324495	0.011419	-0.059752	-0.071170	0.601868	False
30	IC3-IC21	Somatomotor - Default mode	1.663671	0.099415	0.332165	0.216464	0.142239	-0.074226	0.601868	False
19	IC9-IC21	Default mode - Default mode	1.485817	0.140517	0.294747	0.100306	0.034992	-0.065315	0.618590	False
11	IC2-IC3	Visual - Somatomotor	-1.446451	0.151211	-0.287282	0.013902	0.080771	0.066870	0.618590	False

Table 1. Top 10 functional network connectivity edges ranked by statistical significance.

3.2 Group Differences in FNC

Following FDR correction, one functional network connectivity edge remained significant. This edge linked IC9 (Default Mode Network) and IC3 (Somatomotor Network) ($t = 3.70$, $q = 0.016$). Connectivity strength was significantly lower in schizophrenia patients (mean = -0.073) than in healthy controls (mean = 0.088), corresponding to a moderate-to-large effect size (Cohen's $d = 0.74$) (Figure 3).

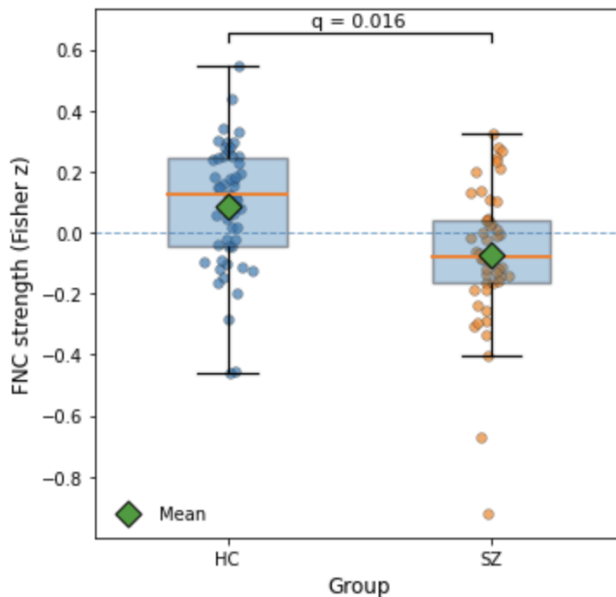


Figure 3. Group differences in functional network connectivity between the Default Mode Network and Somatomotor Network. Boxplots show Fisher z-transformed connectivity values for healthy controls and schizophrenia patients. Schizophrenia patients exhibited significantly

reduced connectivity relative to healthy controls, surviving FDR correction ($q = 0.016$, Cohen's $d = 0.74$).

3.3 Classification Performance

Functional network connectivity features were subsequently used for diagnostic classification. The logistic regression classifier achieved a held-out test AUC of 0.851 (Figure 4). Permutation testing demonstrated that this performance exceeded chance expectations (permutation $p = 0.001$; Figure 5). The mean AUC obtained from permuted labels was approximately 0.50, indicating that the classifier captured meaningful information associated with schizophrenia diagnosis rather than random variation.

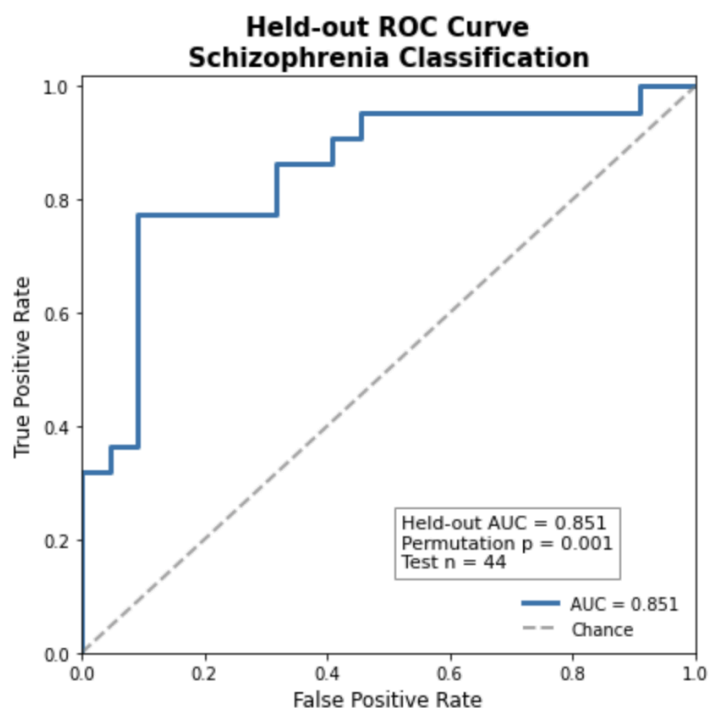


Figure 4. ROC curve for schizophrenia classification using functional network connectivity features. The classifier achieved a held-out test AUC of 0.851.

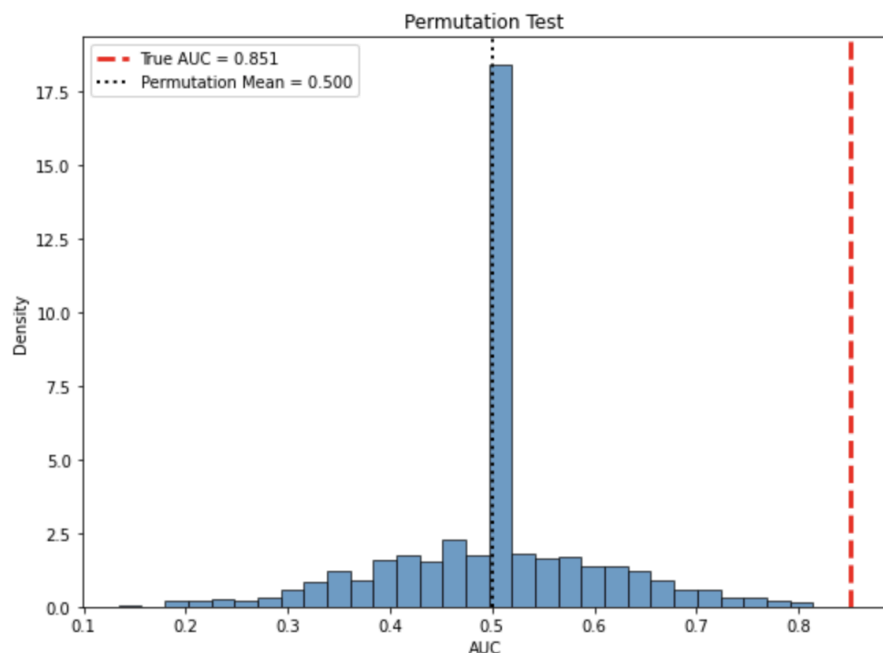


Figure 5. Permutation test results for schizophrenia classification. The red dashed line indicates the true test AUC (0.851), whereas the histogram represents the null distribution generated from 1,000 random label permutations.

4. Discussion

The primary goal of this project was to reproduce a widely used ICA–FNC analysis framework and gain a better understanding of how resting-state fMRI can be used to study schizophrenia. Overall, the pipeline was successfully implemented, including group ICA, network identification, dual regression, functional network connectivity estimation, statistical testing, and machine learning classification. The results suggest that even a relatively simple implementation of this framework is capable of detecting meaningful differences between schizophrenia patients and healthy controls.

One of the most interesting findings was that connectivity between the Default Mode Network (DMN) and the Somatomotor Network was significantly reduced in schizophrenia patients. After correcting for multiple comparisons, this was the only connection that remained significant. Interestingly, only a single edge survived FDR correction. However, the effect size of this connection was moderately large (Cohen's $d \approx 0.74$), suggesting that the difference is meaningful despite the relatively modest sample size.

The biological interpretation of this result remains uncertain. The DMN is commonly associated with internally directed thought and self-referential processing (Raichle et al., 2001; Buckner et al., 2008), whereas the Somatomotor Network is involved in sensory and motor functions (Yeo et al., 2011; Power et al., 2011). One possible interpretation is that schizophrenia involves

altered interactions between internally generated mental activity and externally oriented sensorimotor processing. Although this project cannot directly test such a hypothesis, the finding appears broadly consistent with the dysconnection hypothesis, which proposes that schizophrenia reflects abnormalities in communication among distributed brain systems rather than dysfunction within a single brain region (Friston & Frith, 1995; Stephan et al., 2006; Friston et al., 2016).

The machine learning results were also encouraging. Using FNC features alone, the logistic regression classifier achieved a held-out test AUC of 0.851. Furthermore, permutation testing indicated that this performance was unlikely to arise by chance. While this level of performance should not be interpreted as evidence for a clinically useful diagnostic tool, it suggests that resting-state functional connectivity contains information that is relevant to schizophrenia diagnosis.

The present classification results are broadly consistent with previous work by Arbabshirani et al. (2013), who demonstrated that ICA-derived functional network connectivity features could distinguish schizophrenia patients from healthy controls. Although direct comparison is difficult because of differences in preprocessing, component selection, and validation procedures, the current findings support the general conclusion that large-scale resting-state network interactions contain diagnostically relevant information. More importantly, the result demonstrates that the overall ICA–FNC pipeline is functioning as expected and is able to recover disease-related patterns from the data.

This project also highlighted several limitations. First, only a single dataset was analyzed, making it difficult to assess the generalizability of the findings. Second, the sample size was relatively limited for machine learning applications. Third, some analysis choices, such as ICA component selection and network matching procedures, involved subjective decisions. Different component numbers, network templates, or preprocessing strategies might produce somewhat different results. Finally, the present study examined only static functional connectivity. Because brain networks fluctuate over time, dynamic functional connectivity approaches may reveal additional abnormalities that are not captured by average correlations (Damaraju et al., 2014).

The most valuable outcome of this project was gaining a practical understanding of how modern resting-state fMRI analyses are performed. Before starting this work, concepts such as PCA, ICA, dual regression, and functional network connectivity were mostly ideas encountered in papers. Implementing the full pipeline made it much easier to understand how these methods relate to one another and how large-scale brain networks can be extracted from neuroimaging data. In this sense, the project served both as an investigation of schizophrenia-related network abnormalities and as an introduction to contemporary computational neuroimaging methods.

Future work could investigate whether the observed DMN–Somatomotor finding replicates in other schizophrenia datasets. It would also be interesting to compare static and dynamic connectivity approaches, explore alternative network definitions, and examine relationships between connectivity abnormalities and clinical symptoms. Such analyses may provide a more complete picture of how large-scale brain networks are altered in schizophrenia.

5. Conclusion

In summary, this study successfully reproduced a classical ICA-FNC schizophrenia classification pipeline using the COBRE dataset. Reduced DMN-Somatomotor connectivity emerged as the strongest group difference, and FNC features achieved significant classification performance. These findings support the utility of large-scale functional network interactions as biomarkers of schizophrenia and demonstrate the feasibility of reproducing established neuroimaging pipelines using open datasets.

Citations :

Dysconnection Hypothesis

Friston, K. J., & Frith, C. D. (1995). Schizophrenia: A disconnection syndrome? *Clinical Neuroscience*, 3(2), 89–97.

Stephan, K. E., Friston, K. J., & Frith, C. D. (2006). Dysconnection in schizophrenia: From abnormal synaptic plasticity to failures of self-monitoring. *Schizophrenia Bulletin*, 32(1), 45–55.

Friston, K., Brown, H. R., Siemerkus, J., & Stephan, K. E. (2016). The dysconnection hypothesis (2016). *Schizophrenia Research*, 176(2–3), 83–94. <https://doi.org/10.1016/j.schres.2016.07.014>

rs-fMRI Founding

Raichle, M. E., MacLeod, A. M. A., Snyder, A. Z., Powers, J., Gusnard, D. A., & Shulman, G. L. (2001). A default mode of brain function. *Proceedings of the National Academy of Sciences*, 98(2), 676–682. <https://doi.org/10.1073/pnas.98.2.676>

Fox, M. D., & Raichle, M. E. (2007). Spontaneous fluctuations in brain activity observed with functional magnetic resonance imaging. *Nature Reviews Neuroscience*, 8(9), 700–711. <https://doi.org/10.1038/nrn2201>

Spatial ICA / Group ICA for fMRI

Beckmann, C. F., & Smith, S. M. (2004). Probabilistic independent component analysis for functional magnetic imaging. *IEEE Transactions on Medical Imaging*, 23(2), 137–152. <https://doi.org/10.1109/TMI.2003.822821>

Calhoun, V. D., Adali, T., Pearlson, G. D., & Pekar, J. J. (2001). A method for making group inferences from fMRI data using independent component analysis. *Human Brain Mapping*, 14(3), 140–151.

Damoiseaux, J. S., Rombouts, S. A. R. B., Barkhof, F., Scheltens, P., Stam, C. J., Smith, S. M., & Beckmann, C. F. (2006). Consistent resting-state networks across healthy subjects. *Proceedings of the National Academy of Sciences*, 103(37), 13848–13853. <https://doi.org/10.1073/pnas.0601417103>

FNC (Functional Network Connectivity)

Jafri, M. J., Pearlson, G. D., Stevens, M., & Calhoun, V. D. (2008). A method for functional network connectivity among spatially independent resting-state components in schizophrenia. *NeuroImage*, 39(4), 1666–1681. <https://doi.org/10.1016/j.neuroimage.2007.11.001>

Damaraju, E., Allen, E. A., Belger, A., Ford, J. M., McEwen, S., Mathalon, D. H., Mueller, B. A., Pearlson, G. D., Potkin, S. G., Preda, A., Turner, J. A., Vaidya, J. G., van Erp, T. G. M., & Calhoun, V. D. (2014). *Dynamic functional connectivity analysis reveals transient states of dysconnectivity in schizophrenia. NeuroImage: Clinical*, 5, 298–308. <https://doi.org/10.1016/j.nicl.2014.07.003>

The Seminal Classification Study

Arbabshirani, M. R., Kiehl, K. A., Pearlson, G. D., & Calhoun, V. D. (2013). Classification of schizophrenia patients based on resting-state functional network connectivity. *Frontiers in Neuroscience*, 7, Article 133. <https://doi.org/10.3389/fnins.2013.00133>

INDI / NITRC. *COBRE: Center for Biomedical Research Excellence Schizophrenia Dataset*. International Neuroimaging Data-sharing Initiative (INDI). Available at: http://fcon_1000.projects.nitrc.org/indi/retro/cobre.html [Accessed 2026].

Aine, C. J., Stephen, J. M., Adair, J. C., Knoefel, J., Steiner, E. R., McDonald, C., ..., & Calhoun, V. D. (2017). Multimodal neuroimaging in schizophrenia: Description and dissemination. *Neuroinformatics*, 15(3), 301–315. <https://doi.org/10.1007/s12021-017-9338-9>

Bellec, P. (2017). *NIAK: NeuroImaging Analysis Kit*(Version 1.1.4) [Computer software]. RRID:SCR_002497. <https://github.com/SIMEXP/niak/>. Also available from NITRC: <https://nitrc.org/projects/niak>

Pre-processing

Behzadi, Y., Restom, K., Liau, J., & Liu, T. T. (2007). A component based noise correction method (CompCor) for BOLD and perfusion based fMRI. *NeuroImage*, 37(1), 90–101. <https://doi.org/10.1016/j.neuroimage.2007.04.042>

Cordes, D., Haughton, V. M., Arfanakis, K., Wendt, G. J., Turski, P. A., Moritz, C. H., Quigley, M. A., & Meyerand, M. E. (2001). Frequencies contributing to functional connectivity in the cerebral cortex in resting-state data. *American Journal of Neuroradiology*, 22(7), 1326–1333.

Friston, K. J., Williams, S., Howard, R., Frackowiak, R. S. J., & Turner, R. (1996). Movement-related effects in fMRI time-series. *Magnetic Resonance in Medicine*, 35(3), 346–355. <https://doi.org/10.1002/mrm.1910350312>

Weissenbacher, A., Kasess, C., Gerstl, F., Lanzenberger, R., Moser, E., & Windischberger, C. (2009). Correlations and anticorrelations in resting-state functional connectivity MRI: A quantitative comparison of preprocessing strategies. *NeuroImage*, 47(4), 1408–1416. <https://doi.org/10.1016/j.neuroimage.2009.05.005>

Brain Masking:

Nilearn: Abraham A, Pedregosa F, Eickenberg M, Gervais P, Mueller A, Kossaifi J, Gramfort A, Thirion B, Varoquaux G. *Machine learning for neuroimaging with scikit-learn*. *Frontiers in Neuroinformatics* 2014;8:14.

Time Series Extraction:

Abraham, A., Pedregosa, F., Eickenberg, M., Gervais, P., Mueller, A., Kossaifi, J., Gramfort, A., Thirion, B., & Varoquaux, G. (2014). Machine learning for neuroimaging with scikit-learn. *Frontiers in Neuroinformatics*, 8, 14. <https://doi.org/10.3389/fninf.2014.00014>

Calhoun, V. D., Adali, T., Pearlson, G. D., & Pekar, J. J. (2001). A method for making group inferences from functional MRI data using independent component analysis. *Human Brain Mapping*, 14(3), 140–151. <https://doi.org/10.1002/hbm.1048>

PCA theory

Jolliffe, I. T. (2002). *Principal component analysis* (2nd ed.). Springer.

Calhoun, V. D., Liu, J., & Adali, T. (2009). A review of group ICA for fMRI data and ICA for joint inference of imaging, genetic, and ERP data. *NeuroImage*, 45(1 Suppl.), S163–S172. <https://doi.org/10.1016/j.neuroimage.2008.10.057>

ICA Theory

Hyvärinen, A. (1999). Fast and robust fixed-point algorithms for independent component analysis. *IEEE Transactions on Neural Networks*, 10(3), 626–634. <https://doi.org/10.1109/72.761722>

Hyvärinen, A., & Oja, E. (2000). Independent component analysis: Algorithms and applications. *Neural Networks*, 13(4–5), 411–430. [https://doi.org/10.1016/S0893-6080\(00\)00026-5](https://doi.org/10.1016/S0893-6080(00)00026-5)

Calhoun, V. D., Adali, T., Pearlson, G. D., & Pekar, J. J. (2001). A method for making group inferences from functional MRI data using independent component analysis. *Human Brain Mapping*, 14(3), 140–151. <https://doi.org/10.1002/hbm.1048>

Reconstruction of Group Spatial Maps

Allen, E. A., Erhardt, E. B., Damaraju, E., Gruner, W., Segall, J. M., Silva, R. F., Havlicek, M., Rachakonda, S., Fries, J., Kalyanam, R., Michael, A. M., Caprihan, A., Turner, J. A., Eichele, T., Adelsheim, S., Bryan, A. D., Bustillo, J., Clark, V. P., Feldstein Ewing, S. W., ... Calhoun, V. D. (2011). A baseline for the multivariate comparison of resting-state networks. *Frontiers in Systems Neuroscience*, 5, 2. <https://doi.org/10.3389/fnsys.2011.00002>

Identification of resting state networks

Yeo, B. T. T., Krienen, F. M., Sepulcre, J., Sabuncu, M. R., Lashkari, D., Hollinshead, M., Roffman, J. L., Smoller, J. W., Zöllei, L., Polimeni, J. R., Fischl, B., Liu, H., & Buckner, R. L. (2011). The organization of the human cerebral cortex estimated by intrinsic functional connectivity patterns in 1000 subjects. *Journal of Neurophysiology*, 106(3), 1125–1165. <https://doi.org/10.1152/jn.00338.2011>

Statistical Analysis

Welch BL. *The generalization of Student's problem when several different population variances are involved*. *Biometrika* 1947;34(1–2):28–35.

Benjamini Y, Hochberg Y. *Controlling the false discovery rate: a practical and powerful approach to multiple testing*. *JRSSB* 1995;57(1):289–300.
doi:10.1111/j.2517-6161.1995.tb02031.x

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Machine learning classification:

Ojala M, Garriga GC. *Permutation tests for studying classifier performance*. *JMLR* 2010;11:1833–1863.

Software:

Abraham, A., Pedregosa, F., Eickenberg, M., Gervais, P., Mueller, A., Kossaifi, J., Gramfort, A., Thirion, B., & Varoquaux, G. (2014). Machine learning for neuroimaging with scikit-learn. *Frontiers in Neuroinformatics*, 8, 14. <https://doi.org/10.3389/fninf.2014.00014>

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>

Brett, M., Markiewicz, C. J., Hanke, M., Côté, M.-A., Cipollini, B., McCarthy, P., Ghosh, S., Wassermann, D., Gerhard, S., Larson, E., Lee, G. R., Halchenko, Y. O., Kastman, E., Rokem, A., Madison, C., Morency, F. C., Moloney, B., Goncalves, M., Markello, R., ... Gorgolewski, K. J. (2023). *NiBabel* (Version 5.1.0) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.591597>

Discussion :

Buckner, R. L., Andrews-Hanna, J. R., & Schacter, D. L. (2008). The brain's default network: Anatomy, function, and relevance to disease. *Annals of the New York Academy of Sciences*, 1124, 1–38. <https://doi.org/10.1196/annals.1440.011>